

Mobile Development :

Syllabus



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Outline :

- Objectives & Outcomes
- Course Content
- Why it is important
- Teaching Staff
- Evaluation & Assignments
- Resources
- Ethics & Policies.





Objectives & Outcomes

- Students are expected to learn how to engineer and develop a professional-quality mobile application that addresses a real-world problem in an innovative way.
- Students will learn the various phases of engineering a mobile app product, including planning and idea validation, UI/UX design, implementation, and testing



Objectives & Outcomes

- At the end, students are expected to learn and acquire :
 - Skills to find ideas and turn them into a professional mobile app.
 - Programming Principles & Development for Mobile App.
 - Engineering a Software Product for the Mobile Market.
 - Collaboration with other members to deliver a product.



Why it is important ?

- Because mobile phones are becoming ubiquitous and an integral part of our lives.
- Convenience of digital users to interact with information systems from their smartphones.
- Career Perspectives:
 - Mastering the technologies, concepts, and methodologies for developing mobile applications is an advantage on your CV and will help you secure better job opportunities
 - Most startups being launched recently are based on a mobile-app project.



Why it is important ?

- And more importantly:
 - **Mobile Apps can serve as the product to deliver your AI innovations.**

Content & Planning

Part 1 : Introduction:

- Why Mobile Apps?
- Technology : Native, Web and Hybrid

Part 2 : App Design Issues and Considerations

- Mobile Development Lifecycle Overview
- Architecture, Design and Engineering Considerations
- Usability and User Interaction Design
- Overarching Design Principles and Guidelines

Part 3: Developing the Mobile App

- Techniques, Methodologies for Mobile Application Development
- Mobile Application Development Frameworks
- Persistent Data in Mobile Apps
- Maps and Location in Mobile Apps
- Access to Hardware and Sensors
- Building Mobile Apps Powered by Enterprise Backend
- Secured Data Store and Synchronisation

Part 4 : Testing and Publishing Apps

- Mobile Application Build and Delivery
- Testing Mobile Applications
- App Distribution Through App Stores
- Monetizing Apps



Content & Planning

1. Syllabus + Introduction to Mobile Dev + Dart Programming Essentials.
2. Mobile Apps : Idea, Planning, UX/Design, Marketing & Monetization Methods
3. Flutter : Part 1 Widgets & Events
4. Flutter : Part 2 Interactivity, Navigation & Forms
5. Flutter : Part 3 Network, Async, Future & APIs
6. Flutter : Part 4 Persistence & Storage
7. Flutter : Part 5 - State Management

Content & Planning



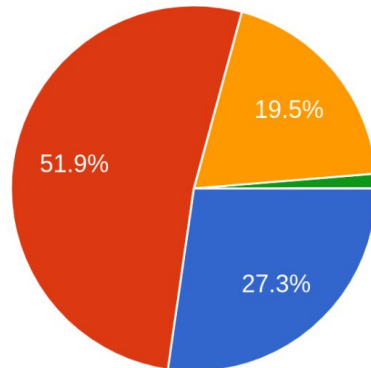
8. Flutter : Part 6 - State Management
9. Integration with Other Services : Backends
10. Integration with Other Services : External Services (Firebase,..)
11. Advanced Topics for Mobile Apps Part1, Background Tasks, Animation, ..
12. Advanced Topics for Mobile Apps Part2, Localization, Native, Packaging, CI/CD
13. Deployment & Testing of Mobile Apps
14. Publishing, Monetizing and Promoting Apps

Why it is important ?

- Feedback from prev students (2022/2023)

I feel that this module is :

77 responses



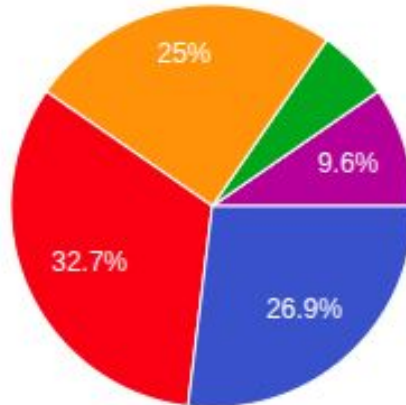
- Very Important
- Important
- Normal
- Not important
- Better if it is removed

Why it is important ?

- Feedback from prev students (2024/2025)

I feel that this module is :

52 responses



- Very Important
- Important
- Normal
- Not important
- Better if it is removed

Teaching Staff

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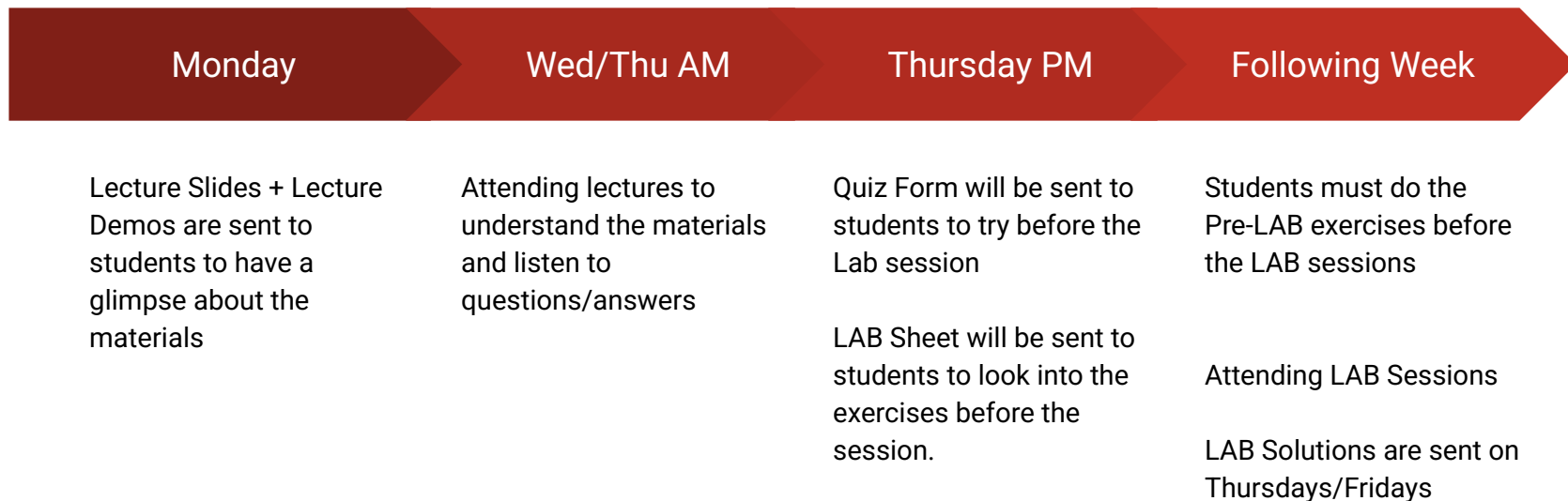
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Learning Process





Evaluation & Assignments

- Exam : 60%
 - Continuous Evaluation : 40%
 - Standard Mark : **20 Points**
 - **Group** Mini Project : **8 points**
 - Mid-Term Exam : Individual Coding Exercise : **7 points**
 - Online Quizzes : **2 points**
 - Completion of Lab-Exercises & Staff Appreciation : **3 points**
 - Bonus/Extra Points: **5 Points**
 - **Individual** Challenge Exercises : **3 points** (Optional)
 - Midterm Solution to Optional Exercises : **2 points** (Optional)
 - Penalty of Non-Justified Absence : **-5 points**
 - Late Submission :
 - Not Accepted. If you submit by Email, it will be deleted automatically.
- 



Quizzes (8 points)

- Quizzes will be available either :
 - After the lecture and closed at the end of the week.
 - During the lecture only.
 - Only for the students who attended the lecture(s)
- Missing three **quizzes**, zero mark will be given directly and all other submissions will not be considered.
- There will be only 8 quizzes
- **Being absent in 5 lectures, ZERO mark will be given.**



Quizzes (2 points)

- Online Quizzes should preferably be done before attending the tutorial sessions.
- Quizzes include easy/medium M.C.Questions aimed to enforce your understanding for the concepts discussed during the lecture.
- PLEASE : Don't rush to answer the quiz, take your time, **search for the solutions on the web** and submit your answers when you are certain.
- The quiz will be open during the week of the lecture. (To ensure students will not procrastinate doing their revision and homework)

Group Mini Project (8 points)

- ***Team Creation :***

- Teams of 2 or 3 or 4 members will be created (**No teams of 5 members or ONE**) .
- Team members should be from the same group (or under the same LAB Lecturer)
- You are **not** allowed to change your team to another team when the topic is assigned.

- ***Team Leader :***

- Up to the team to have a democratic process to select a leader or any other process they like to invent .
- All correspondence, reporting and communication must be done by the leader

Group Mini Project (8 points)

- Project Topic :
 - The team members must collaborate to come up with an idea for a project to implement.
 - Provided it is validated by a staff
 - The project topics must :
 - Feasible within the resources you have
 - Will be used by real people .
 - It solves an existing problem
 - **Not over ambitious.**
 - **Technology :** We teach Flutter, we expect strictly Flutter to be used.

Group Mini Project (8 points)

- Process for Project Topic :
 - Submission of Project Title only
 - *Period : 5 October – 12 October 2024*
 - *It will be reviewed by the LAB Lecturer*
 - *If rejected, you need to either : Submit new project title or re-adjust.*
 - *If accepted, proceed directly to work on the specification document*
 - Submission of the Specification & Planning document
 - *Deadline : 19 October 2024*
 - *You tell us what functionalities we are expecting to see on your mobile app.*

Group Mini Project (8 points)

- Process for Project Topic :

- Submission of Project Title only

- *Period : 5 October*

- *It will be reviewed*

- *If rejected, you can*

- *If accepted, proceed*

- Submission of the S

- *Deadline : 19 October*

- *You tell us what*

**Teams are encouraged to seek ideas away from
the education arena**

**App for uni, App for BAC, App for Schools, App for
students to students...**

.(Please, move away from your comfort zone

Group Mini Project (8 points)

- Marking Per Student or Per Team :
 - We will give one Mark per Team.
 - You work all a a team, you either all win or all lose.
 - If other students are not contributing ? it is not fair that he/she gets the same mark as a working person ? :
 - There will be an individual evaluation per members.
 - Students not attending team meetings, Lab sessions, lectures will be penalized.

Group Mini Project (8 points)



- Marking Schema :
 - Continuous Delivery of Sprint Increments: **2 points**
 - Final Delivery: **6 points**
 - Functional Software : 1.5 points (Non functioning software $\Rightarrow$ opts)
 - Coding, Implementation and Architecture : 1 points
 - There are considerable real* users of your software : 1 points
 - Non-Functional Requirements : 0.5 points
 - UX Design : 0.5 points
 - Existence of Backend : 0.5 point
 - Documentation (User & Technical): 0.5 points
 - Demonstration : 0.5 points
 - Published on Google Play : +1.5 Point Extra

**Real users means not fake, bots, or ENSIA students or relatives*

Group Mini Project (20 points)

- Exceptional Project :
 - The team developing a professional software having an exceptional impact
 - Considerable* number of **active real users**.
 - Fully functional
 - Professional Branding and Packaging
 - Students of the selected team, will get the full mark for :
 - All the quizzes (Provided they don't miss three quizzes)
 - The group project
 - Only one team will be selected

Important Dates



- **05 October** : Creation of Teams
- **12 October** : Deadline for Proposing Project Titles
- **23 October** : Deadline for Submitting Project Specifications
- **09 November** : Submission of the Design for UI/UX
- **06 December** : Release of MVP
- **14 December** : Release of Sprint 1 for Feedback
- **10 January** : Final Submission of the project deliverables
- **Final Week** : Project Validation during Lab Sessions.

Mid-Term Coding Exercise (7 points)

- Individual Exercise
 - Students would be given a topic (or set of topics to choose from)
 - Based on what they learned in lectures and lab sessions, they will develop a mobile application with a minimal set of functionalities.
 - It will be an individual exercise. Books, PCs, online resources are permitted.
 - Evaluation Schema :
 - Per Task Given.
- Bonus Questions (**Extra 2 points**) :
 - Students may asked to submit Solutions to certain Optional Exercises.



Exams & Midterms

- There will be Questions in the Midterm or the Exam being :
 - Taken from the Optional Exercises
 - Asked inside the lecture
- For the marking procedure, to ensure fairness, each exercise/question is marked by the **same lecturer** for all students.
- If you are incorrect for a given answer, zero mark should be given straight away. However, we may subjectively give some points in appreciation to the effort made by the student. But, we are not required nor you have the right to claim points for incorrect or partially correct answers.

Challenge Questions

(3 points of extra credit)

- Meant for you to learn how to solve difficult problems that you may face in the future and you can not find their solutions on the web freely.
- Solutions for the challenge Exercises will be submitted via Google Classroom when the Assignments are created. **(It will be created without prior notice with only a one day deadline or subjected to quota)**
- Solving the exercises is optional, but you are encouraged to do so.



Lab Sessions - TP

- Exercises of Prefix :
 - **C** : to be solved in class where you will be given some time to try to solve them.
 - **A : Assignments/homework** that you need to solve within a week deadline. You need to submit your solution via Google Classroom
 - **P** : Are optional exercises to be done at home after class. Discuss with your colleagues about the possible solutions. Feel free to bring your solutions to the lecturer to discuss in class when convenient.
- Note that all **P exercises** are **potential exam** questions that they **will** be included into the final exams.



Lab Sessions - TP

- How it works:
 - We will go through each student individually checking :
 - *They have done the pre-lab exercises.*
 - *Progressed well with the previous exercises*
 - Students proceed to work **individually** on the exercises : C1 and C2
 - **Pause/Break of 10 Minutes**
 - Students should resume to work on C3
 - 30 minutes may be allocated for students to discuss how to use the new concepts towards their project.
 - Students need to submit screenshots of their solution using the form provided to ensure completion of all in-class exercises.



Lab Sessions - TP

- TP Sessions will serve as :
 - To practice and master concepts introduced during the lecture
 - Meeting sessions where teams brief their lecturer about their progress.
- **Important Note :**
 - Lab Session time should never be wasted to install software or configure them.
 - Lab instructors would offer guidance and recommendations on coding strategies, principles, tips and explanations. But don't expect us to write the code or **debug it**. (As it can be time consuming and unfair to other students seeking legitimate assistance)
- If you have done all exercises yourself before the lab ?

Absence & Penalties

- LAB Sessions :
 - Students being absent more than allowed, law will be strictly applied and students will be excluded.
 - Students need to :
 - coordinate with the **administration** and LAB staff about their absences.
 - Use the form to notify us too: <https://forms.gle/4cWnVRXhKGpFbBYFA>
 - 5 lab absences (Being absent for 5 weeks) :
 - CC mark of **zero will be given** directly regardless of the project, midterm...
 - Attendance File :

https://docs.google.com/spreadsheets/d/1Vpvftb2jQLiOVDO4A-gO-qR_NKiJUgqyBy5-p7whK5o/edit?gid=94491315#gid=94491315

Absence & Penalties

- Lecture Sessions

- Students are **advised** to attend all lectures.
- If you miss the lecture, don't expect LAB staff to repeat lecture materials.
- Students who are were absent for a **valid reason**, should inform us via the form <https://forms.gle/4cWnVRXhKGpFbBYFA> (no need to inform the administration)
- Attendance File :
<https://docs.google.com/spreadsheets/d/1tXJsZrAaSm2zwlFSUvAEXngWsFcr23au19GGZwXoom8/edit?gid=94491315#gid=94491315>
- Reasons not accepted:
 - Driving license exam.
 - Didn't wake up.
 - Have a job somewhere else.



Absence & Penalties

- Being late :
 - Students coming 10 minutes late, will be allowed in class but they will be considered **ABSENT**
- Absence Penalty Points :
 - **-0.25 Point per Non-Justified Absence** (LAB Sessions OR Lectures)
 - Lab session = one session of 90 minutes
 - A max of **5 points** can be deducted from the Continuous Mark.
- More than **>= 5** absences :
 - All Bonus Points (Challenge, Midterm....) are disregarded.
 -



Attending Lectures

- Can the lectures be substituted by Youtube videos given by Top Scientists from MIT and Stanford ?
- Are lectures about telling you what's inside the book that you can read on your own ?
- Deliberate missing of lectures :
 - Would contribute to create a lazy atmosphere among the students
 - Accumulation of pressure and stress before the exam.
- Of course, You **may** learn a lot on your own without attending lectures, but for 7 modules within a semester, you are putting yourself at a risk.



Exceptional Cases

- Exceptional Progress : Mid-term vs. Exam Marks :
 - Students who scored below 10 in the midterm whilst their score above ($>$) 15 in the exam, their midterm mark will be the same as the exam.
- Exceptional Projects :
 - Students who come up with an Exceptional Project with a true impact, they will get the full mark for the continuous mark.
- Exceptional Cases of Justified Absences:
 - Students who have missed considerable number of sessions, please always contact us for assistance when needed to catch up.

Contacting us

- Justifying an Absence : (don't send an email) :
 - <https://forms.gle/NuMFkxn8FG4TAuNx7>
- Asking a question or Feedback related to the lecture or Lab:
 - Send us an email. Be brief please



Policies

- The use of **Mobile Phones inside the classroom is prohibited for Chat and non-educational purposes**. Disciplinary measures would be taken as it is considered as a disrespect to the lecturers and to the school.
- Chewing Gum/Coffees/Slippers/... are strictly prohibited inside the classrooms
- Penalty for Late Submissions will be applied strictly. Any submission that we cannot open or unzip, zero mark will be given directly. Submission by email email is never acceptable.
- If you need assistance or have any questions, always speak directly to the lecturer, lab staff and make use of the weekly office hours.

Policies :

ChatGPT/Claude

- ChatGPT is totally forbidden in the MobileDEV Course
 - Most exercises can be solved by ChatGPT, but the aim is teach you how to become a good problem solver, inventor and innovator by coming up with solutions for problems/bugs/...
 - Most questions can be answered/solved by ChatGPT, students need to try and think to come up with answers themselves by trying, making mistakes.
 - We would be **very angry** seeing you opening ChatGPT in LAB Sessions asking computers to replace your humanly brain.